Calculus A(1)
12/5

- We have a class on 12/10 19:20 - 21:45
- Final exam
  12/30 9:00am - 11:00 am
  Online.

Last time :

We saw the definite integral

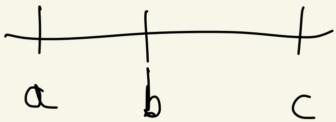
$$\int_a^b f(x)\,dx \qquad f : [a,b] \to \mathbb{R} \text{ integrable}$$

$$\int_a^b f(x)\,dx = \lim_{\substack{P \text{ partition of } [a,b] \\ \|P\| \to 0}} S_P$$

We have seen: C^0 fcts on $[a,b]$ are integrable

Some basic properties of $\int$:

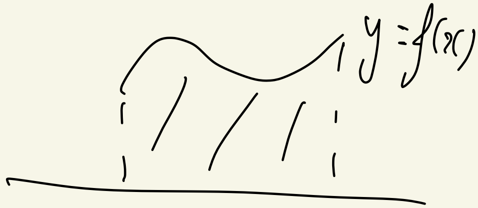
e.g. • $\displaystyle\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$



• Min-Max inequality:

$$(b-a)\min_{[a,b]}(f) \leq \int_a^b f(x)\,dx \leq (b-a)\max_{[a,b]}(f)$$

Back to the area under the graph of $y = f(x)$

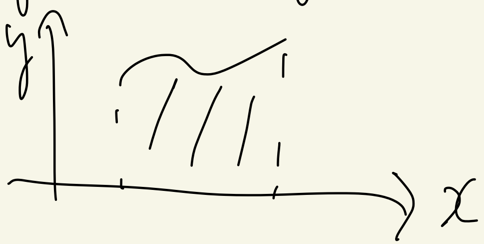


Def (area below a curve)

Let $f : [a,b] \to \mathbb{R}$ s.t $\boxed{f \geq 0}$ (ie. $\forall x \in [a,b], f(x) \geq 0$)

The area under the curve $y = f(x)$ over $[a,b]$ is $\int_a^b f(x)\,dx$.

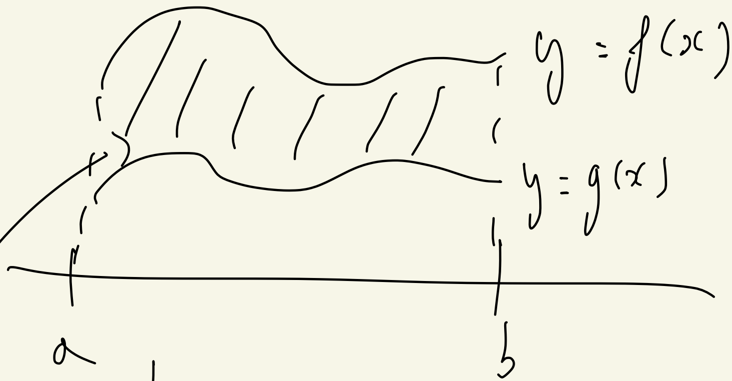
Rem.: This was motivated by Riemann sums, ie. by approximation by small rectangles. This is our first rigorous definition of the area.



We can generalize this for the area between two curves

Let $f, g : [a,b] \longrightarrow \mathbb{R}$

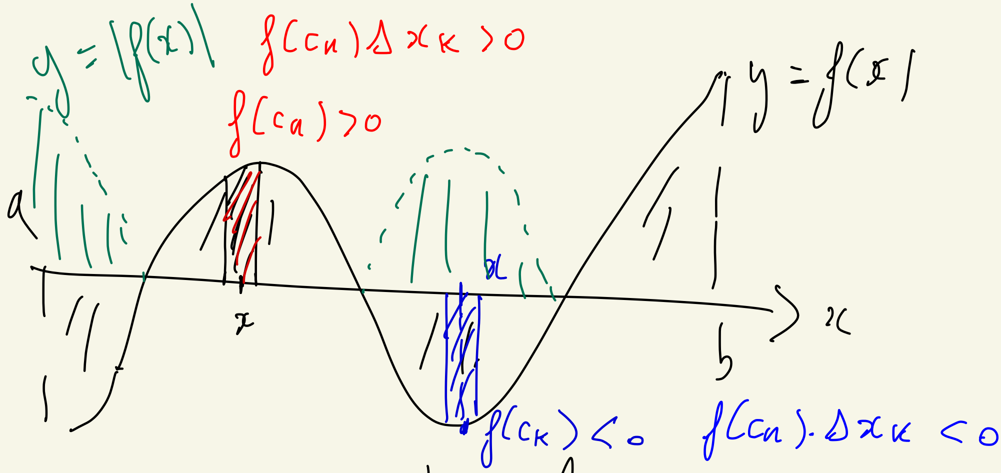
s.t $\forall x \in [a,b]$, $\boxed{f(x) \geq g(x)}$



$$A :\overset{\text{def}}{=} \int_a^b \left(f(x) - g(x)\right) dx$$

$$= \int_a^b f(x) dx - \int_a^b g(x) dx$$

What about functions $f : [a,b] \longrightarrow \mathbb{R}$ taking both positive and negative values?



What is the <u>total area</u>?

$$A := \int_a^b |f(x)|\,dx$$

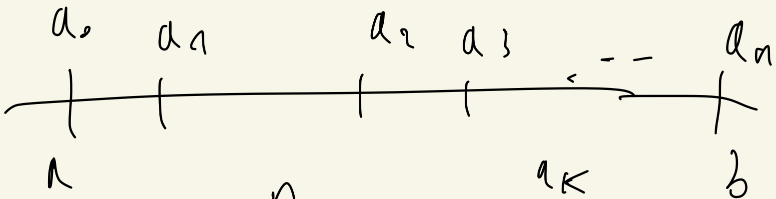
(if $|f|$ is integrable on $[a,b]$)

In practice, to compute A (= the total area), we try to subdivide $[a,b]$ into a finite union of subintervals on which

f has constant sign.

$$[a,b] = [a_0, a_1] \cup [a_1, a_2] \cup \dots \cup [a_{n-1}, a_n]$$

$$a_n = b$$

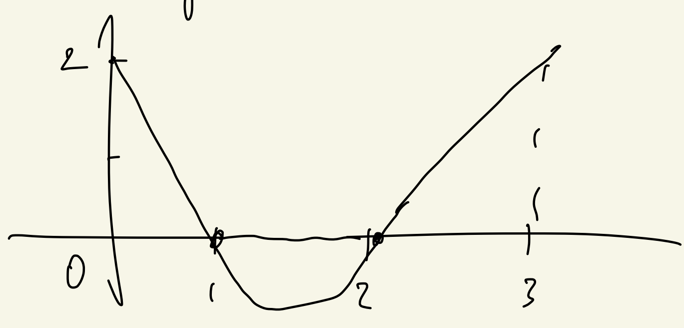


$$A = \sum_{k=1}^{n} \pm \int_{a_{k-1}}^{a_k} f(x)\,dx$$

sign of f on $[a_{k-1}, a_n]$

Ex: $f : [0,3] \longrightarrow \mathbb{R}$

$$f(x) = (x-1)(x-2)$$

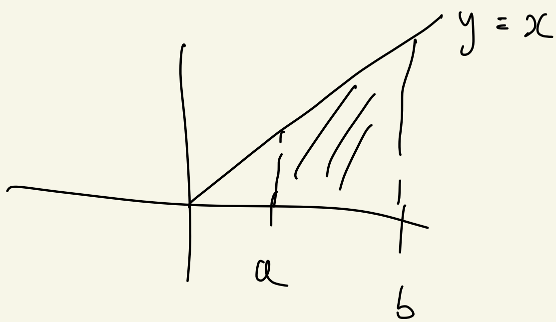


$$A = \int_0^1 f(x)\,dx - \int_1^2 f(x)\,dx + \int_2^3 f(x)\,dx$$

easy to compute (see later)

Ex of area computation :

Find the area between $y = x$ and the x-axis on $[a,b]$ if $a, b \geq 0$



From classical geometry:

$$(b-a)\frac{(b+a)}{2} = \frac{b^2-a^2}{2}.$$

Using definite integrals:

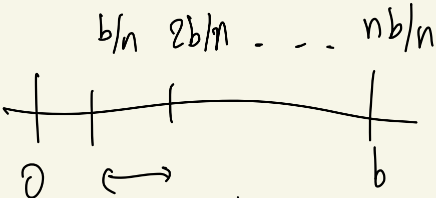
$$\int_a^b x\,dx = \int_0^b x\,dx + \int_a^0 x\,dx$$

$$= \int_0^b x\,dx - \int_0^a x\,dx.$$

Just need to compute

$$\int_0^b x\,dx = \lim_{n\to+\infty} \sum_{k=0}^{n-1} \Delta x f\left(\frac{bk}{n}\right)$$

$$\Delta x = \frac{b}{n}$$



$\Delta x = \frac{b}{n}$

$$= \lim_{n\to+\infty} \sum_{k=0}^{n-1} \frac{b}{n} \cdot \frac{bk}{n}$$

$$= \lim_{n\to+\infty} \frac{b^2}{n^2}\left(\sum_{k=0}^{n-1} k\right)$$

$$= \lim_{n\to+\infty} \frac{b^2}{n^2} \frac{n(n-1)}{2}$$

$$= \frac{b^2}{2}$$

Conclusion: $\displaystyle\int_0^b x\,dx = \frac{b^2}{2}$.

So $\displaystyle A = \frac{b^2}{2} - \frac{a^2}{2} = \frac{b^2 - a^2}{2}$.

$\longrightarrow$ This coincides with the expected result from classical geometry.

Average value / Mean value

Def: Let $f : [a,b] \longrightarrow \mathbb{R}$ integrable

We define $\displaystyle \underset{[a,b]}{\mathrm{av}}(f) = \frac{1}{b-a} \int_a^b f(x)\,dx$

Called the average/mean value of f on $[a,b]$

Rem: Motivation:

$c_1, c_2, \dots, c_n \in [a,b]$

$c_k \in [x_{k-1}, x_k]$ $\qquad P = \{x_0, \dots, x_n\}$ Partition

$$\frac{f(c_1) + \dots + f(c_n)}{n}$$

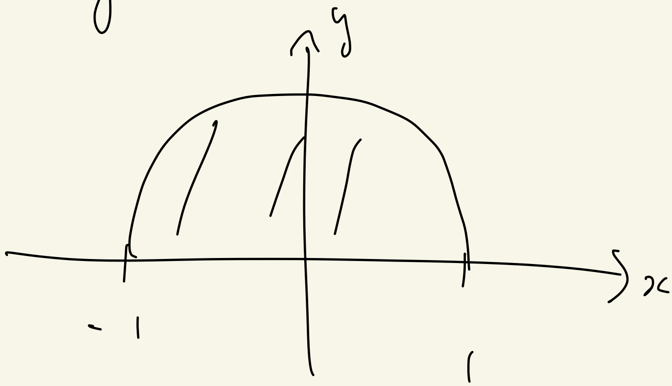
$$= \frac{1}{b-a} \sum \Delta x_k \cdot f(c_k)$$

$$\xrightarrow{n \to +\infty} \frac{1}{b-a} \int_a^b f(x)\,dx$$

Rem: if f has a max and a min on $[a,b]$, then

$$\min(f) \leq av(f) \leq \max(f)$$

Ex: $f(x) = \sqrt{1-x^2}$ on $[-1, 1]$



$$A = \int_{-1}^{1} f(x)\,dx = \frac{\pi}{2}$$

So $av(f) = \dfrac{A}{1-(-1)} = \dfrac{A}{2} = \dfrac{\pi}{4}$

The fundamental theorem of Calculus

We would like to compute definite integrals. So far we have only seen Riemann sums: not convenient.

In this section, we relate definite integrals to indefinite integrals, i.e to antiderivatives. $f : I \to \mathbb{R}$

$$\int f(x)\,dx = F(x) + C$$

where F is any antiderivative of f.

Our main result is called the Fundamental Theorem of Calculus (FTC).

We state two versions.

<u>Theorem</u> (FTC, Version 1)

Let $f : [a,b] \to \mathbb{R}$ C^0.

Then $F(x) = \int_a^x f(t)\,dt$

is differentiable on $[a,b]$ and

$\forall x \in [a,b]$, $F'(x) = f(x)$

In particular, f has an antiderivative on $[a,b]$.

Let us first give an intuitive proof.

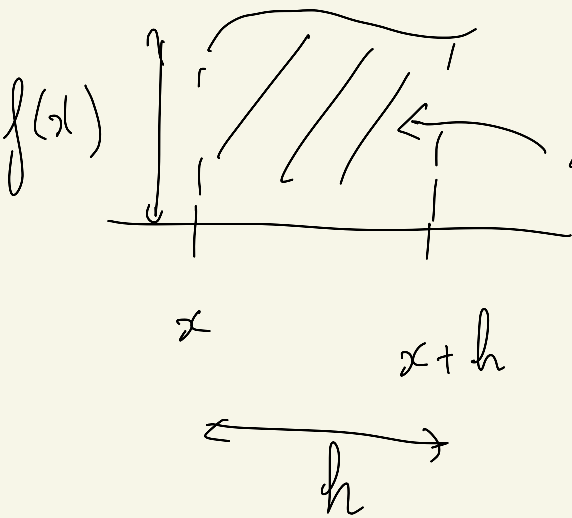
Let $x \in [a,b]$ and $h \in \mathbb{R}$ "small"

$$\frac{F(x+h) - F(x)}{h} = ?$$

$$F(x+h) - F(x) = \int_a^{x+h} f(t)\,dt - \int_a^{x} f(t)\,dt$$

$$= \int_a^{x+h} f(t)\,dt + \int_x^{a} f(t)\,dt$$

$$F(x+h) - F(x) = \int_x^{x+h} f(t)\,dt$$



$A \simeq h \cdot f(x)$

if h very small

So $\quad F(x+h) - F(x) \simeq h \cdot f(x)$

So $\quad \dfrac{F(x+h) - F(x)}{h} \simeq f(x)$

So we <u>expect</u>: $\lim_{h \to 0} \dfrac{F(x+h) - F(x)}{h}$

$$= f(x)$$

i.e. $F'(x) = f(x)$.

<u>Ex</u>:

① $\dfrac{d}{dx}\left(\displaystyle\int_a^x \sin(t)\,dt\right) \overset{\underline{FTC}}{=} \sin(x)$

② What is the antiderivative F of $f(x) = \dfrac{1}{1+x^2}$ s.t. $F(0) = 1$?

$$F(x) = \int_0^x \frac{1}{1+t^2}\,dt + C$$

$$F(0) = 1 \rightsquigarrow C = 1$$

because $\displaystyle\int_0^0 \frac{1}{1+t^2}\,dt = 0$

So $\displaystyle F(x) = \int_0^x \frac{1}{1+t^2}\,dt + 1$

<u>Rk</u>: We will see later that

$F(x) = \arctan(x) + 1$

3) $\displaystyle \frac{d}{dx}\left(\int_1^{x^2} \sin(t)\,dt\right) = ?$

⚠ it is <u>not</u> $\sin(x^2)$

because we have x^2 as an integral boundary, not x.
The FTC only works for x.

If $F(x) = \int_1^x \sin(t)\,dt$

then $F'(x) = \sin(x)$

We want $\frac{d}{dx} F(x^2)$.

Chain rule: $\frac{d}{dx}\left(F(x^2)\right) = 2x \cdot F'(x^2)$

$= 2x \sin(x^2)$

Hence $\frac{d}{dx}\left(\int_1^{x^2} \sin(t)\,dt\right) = 2x \sin(x^2)$

Let us now give a rigorous proof of the FTC (version 1). We shall need the following variant of the MVT for definite integrals

Theorem (MVT for definite integrals)

Let $f : [a,b] \to \mathbb{R}$ $\quad \underline{\underline{C^0}}$

Then $\exists\, c \in [a,b]$ s.t

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

$$\left(= \operatorname{av}_{[a,b]}(f) \right)$$

Proof: We have

$$\min(f) \leq \operatorname{av}(f) \leq \max(f)$$

exist because f is C^0

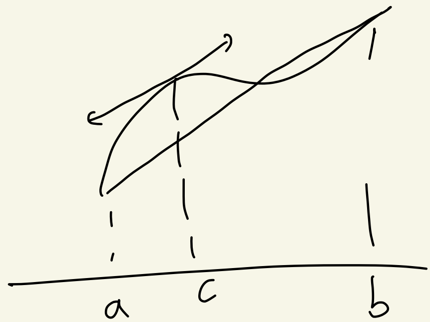
By the IVT, (since f is C^0)

$\exists\, c \in [a,b]$ s.t

$$f(c) = \operatorname{av}(f). \quad \boxtimes$$

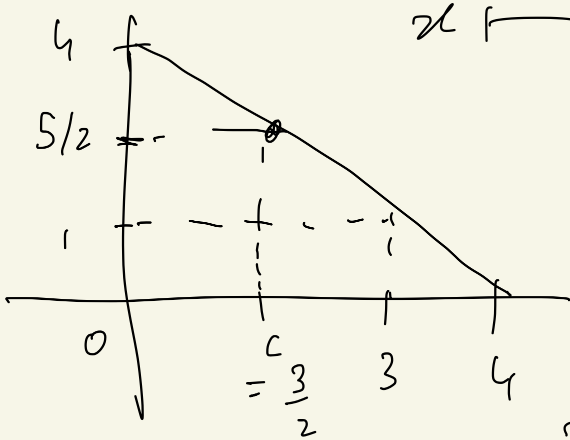
Rem: We will a bit later the relation with the MVT for derivatives:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Ex: $f : [0, 3] \longrightarrow \mathbb{R}$

$x \longmapsto 4 - x$



$$\underset{[0,3]}{Av}(f) = \frac{1}{3-0} \int_0^3 (4 - x)\, dx$$

$$= \frac{1}{3} \left(\int_0^3 4\, dx - \int_0^3 x\, dx \right)$$

$$= \frac{1}{3}\cdot\left(4\cdot 3 - \frac{3^2}{2}\right)$$

$$= \frac{1}{3}\left(\frac{24-9}{2}\right) = \frac{15}{6} = \frac{5}{2}.$$

So MVT for definite integrals tells us: $\exists\, c \in [0,3]$ s.t

$$f(c) = \frac{5}{2}$$

$$c = \frac{3}{2}.$$

<u>Rem</u>: In general, it may not be possible to find c explicitely.

<u>Corollary</u> of the MVT: $f : [a,b] \to \mathbb{R}$ C^0

if $\int_a^b f(x)\,dx = 0$ then $\exists\, c \in [a,b]$ s.t $f(c) = 0$

$\llcorner$ Because : $\underset{[a,b]}{\mathrm{av}}(f) = 0 = f(c)$.

Let us now prove the FTC.

$f : [a,b] \to \mathbb{R}$ C^0

$$F(x) = \int_a^x f(t)\,dt$$

Let h small enough.

$$\frac{F(x+h) - F(x)}{h} \underset{\substack{\text{computed} \\ \text{before}}}{=} \frac{\int_x^{x+h} f(t)\,dt}{h}$$

By the MVT, $\exists\, c \in [x, x+h]$

s.t $f(c) = \frac{1}{h} \cdot \int_x^{x+h} f(t)\,dt$

$$= \operatorname*{av}_{[x, x+h]}(f)$$

A priori, c depends on x and h.

But $c \to x$ when $h \to 0$

Because f is C^0 at x, $f(c) \xrightarrow[h \to 0]{} f(x)$

So $$\frac{F(x+h) - F(x)}{h} = f(c) \xrightarrow[h \to 0]{} f(x)$$

So we have proved F is differentiable at x and

$$F'(x) = f(x).$$

$\boxtimes$

<u>Theorem</u> (FTC, Version 2)

Let $f : [a,b] \to \mathbb{R}$ $\underline{\underline{C^0}}$.

Then f has an antiderivative

$F : [a, b] \longrightarrow \mathbb{R}$

and $\displaystyle\int_a^b f(x)\,dx = F(b) - F(a)$

Rem: if G is another antiderivative of f, then $G(x) = F(x) + C$ for some $C \in \mathbb{R}$

So $G(b) - G(a) = (F(b) + C) - (F(a) + C)$

$= F(b) - F(a)$

Proof: let $F(x) = \displaystyle\int_a^x f(t)\,dt$.

By the FTC (version 1), we know that $F'(x) = f(x)$.

So F is an antiderivative of f.

We have $F(a) = \int_a^a f(t)\,dt = 0$

$$F(b) = \int_a^b f(t)\,dt$$

So $$F(b) - F(b) = \int_a^b f(t)\,dt \quad \boxtimes$$

Notation : $F(b) - F(a) =: \left[F(t)\right]_a^b$

$$=: F(t)\Big]_a^b$$

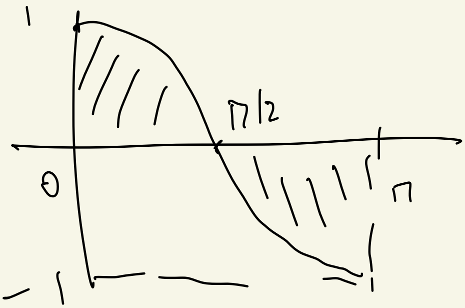
Conclusion : To compute $\int_a^b f(t)\,dt$ where f is C^0,

1. Find an antiderivative F of f.
2. Compute $\left[F(x)\right]_a^b$.

Examples :

① $$\int_0^{\pi} \cos(x)\,dx = \left[F(x) \right]_0^{\pi}$$

$$F(x) = \sin(x) + C$$



$$= \sin(\pi) - \sin(0)$$

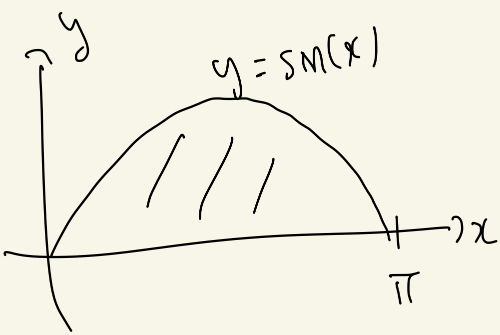
$$= 0 - 0 = 0.$$

② $$\int_0^{\pi} \sin(x)\,dx = \left[-\cos(x) \right]_0^{\pi}$$

$$= -\cos(\pi) - (-\cos(0))$$

$$= -(-1) - (-1)$$

$$= 2$$



$$\Longrightarrow \underset{[0,\pi]}{\mathrm{Av}}(\sin) = \frac{2}{\pi}$$

③ Let $n \in \mathbb{Z}$, $n \neq -1$

$$\int_a^b x^n \, dx = \left[\frac{1}{n+1} x^{n+1} \right]_a^b$$

(if $n < 0$, assume $a \cdot b > 0$)

$$= \frac{b^{n+1}}{n+1} - \frac{a^{n+1}}{n+1}$$

(if $n = 1$: we get $\int_a^b x \, dx = \frac{b^2}{2} - \frac{a^2}{2}$)

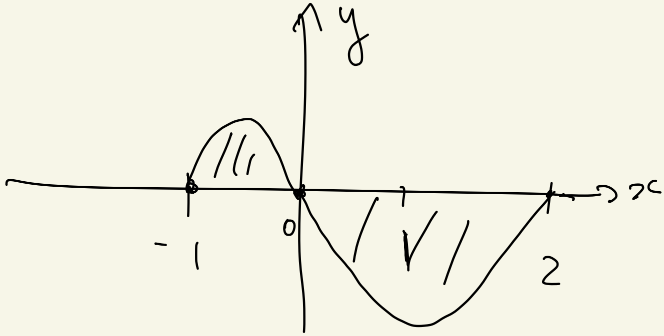
④

$$\int_0^{\pi/4} \sec(x)^2 \, dx = \left[\tan(x) \right]_0^{\pi/4}$$

$$= \underbrace{\tan(\pi/4)}_{= 1} - \underbrace{\tan(0)}_{= 0}$$

$$= 1$$

⑤ Find the (total) area between $y = x^3 - x^2 - 2x$ and the x-axis on $[-1, 2]$



$$\left(\Delta = b^2 - 4ac \qquad x = \frac{-b \pm \sqrt{\Delta}}{2a} \right)$$

$$f(x) = x^3 - x^2 - 2x = x(x^2 - x - 2)$$
$$= x(x-2)(x+1)$$

$$A = \int_{-1}^{0} f(x)\,dx - \int_{0}^{2} f(x)\,dx$$

Let $F(x)$ be an antiderivative of $f(x)$. We have

$$F(x) = \frac{x^4}{4} - \frac{x^3}{3} - x^2$$

So $A \overset{FTC}{=} \left[F(x) \right]_{-1}^{0} - \left[F(x) \right]_{0}^{2}$

$$A = \big(F(0) - F(-1)\big) - \big(F(2) - F(0)\big)$$

$$\boxed{A = 2F(0) - F(-1) - F(2)}$$

$$= \cdots = A = \frac{37}{12}.$$

Rem: Link between the MVT for derivatives and the MVT for integrals.

$$\frac{g(b) - g(a)}{b-a} = g'(c) \qquad \text{MVT for derivatives}$$

$$\frac{1}{b-a}\int_a^b f(t)\,dt = f(c) \qquad \text{MVT for integrals}$$

for some $c \in [a,b]$

Let $f(x) = g'(x)$. Assume g' is C^0.

We get : $$\underbrace{\frac{1}{b-a}\int_a^b g'(x)\,dx}_{\overset{\text{FTC}}{=}\ \frac{g(b)-g(a)}{b-a}} = g'(c)$$

So the MVT for derivatives for g follows from the MVT for definite integrals for g' (if g' is C^0).

The substitution rule

(= change of variable in the integral)

So far, our main technique to compute $\int_a^b f(x)\,dx$ (for f C^0) is to find an antiderivative F of f. (F.T.C).

Not easy (can be impossible) to "find" F in general.

We now give a rule which can help to compute $\int_a^b f(x)\,dx$ or $\int f(x)\,dx$ when we don't

See a priori easily what F is.

We start with an example.

Compute $\int \cos(x) \cdot \sin(x)^2 \, dx$

<u>Trick</u>: Let $u = \sin(x)$

$$du \overset{\text{def}}{=} u'(x)\,dx = \cos(x)\,dx$$

differential of u.

Using this notation, we get

$$\cos(x) \cdot \sin(x)^2 \, dx$$

$$= u^2 \, du$$

So we get $\int \cos(x) \sin(x)^2 \, dx$

$$= \int u^2 \, du$$

$$= \frac{u^3}{3} + C$$

$$= \frac{\sin(x)^3}{3} + c$$

So it suggests:

$$\int \cos(x)\sin(x)^2\,dx = \frac{\sin(x)^3}{3} + c$$

This is indeed true!

This trick works in general.

Examples :

① $$\int \sqrt{1+y^3}\, y^2\,dy = ?$$

Let $u = 1 + y^3$

$$du = 3y^2\,dy \quad \left(\overset{\text{def}}{=} u'(y)\,dy \right)$$

$$\sqrt{1+y^3}\, y^2\,dy = \sqrt{u}\,\frac{du}{3}$$

So $\int \sqrt{1+y^3}\, y^2 dy = \frac{1}{3}\int \sqrt{u}\, du$

$\alpha \neq -1$

$u^\alpha \rightsquigarrow \frac{1}{\alpha+1} u^{\alpha+1}$

$$= \frac{1}{3} \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$$

$$= \frac{1}{3} \frac{u^{3/2}}{3/2} + C$$

$$= \frac{2}{9} \cdot u^{3/2} + C$$

$$\int \sqrt{1+y^3}\, y^2 dy = \frac{2}{9}(1+y^3)^{3/2} + C$$

$\rightsquigarrow$ it is indeed true!

(2) $\int \frac{\cos(x)}{\sin(x)^2} dx = ?$

$u = \sin(x)$. $du = \cos(x)dx$

$$\int \frac{\cos(x)}{\sin(x)^2}\, dx = \int \frac{du}{u^2}$$

$$= -\frac{1}{u} + C$$

$$= -\frac{1}{\sin(x)} + C$$

This is true!